Telco Customer Churn Analysis

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Executive Summary

The telecom company is experiencing a 26.5% annual customer churn rate, resulting in significant revenue loss. This analysis uses machine learning to identify customers at risk of churning before they leave, enabling proactive retention efforts.

Key Finding: An XGBoost model achieves 85% recall in identifying churners, allowing the retention team to intercept high-risk customers and offer targeted retention deals.

Business Impact: For a 5,000-customer base, this model could save approximately KES 340,000 per month in lost revenue.

Problem Statement

Customer churn directly impacts profitability in the telecom industry:
- Acquiring a new customer costs 5-7x more than retaining an existing one
- Current churn rate: 26.5% (~1,865 customers per year in a 5,000-customer base)
- Without early warning, the company cannot proactively intervene
- Solution: Build a predictive model to flag at-risk customers for targeted retention

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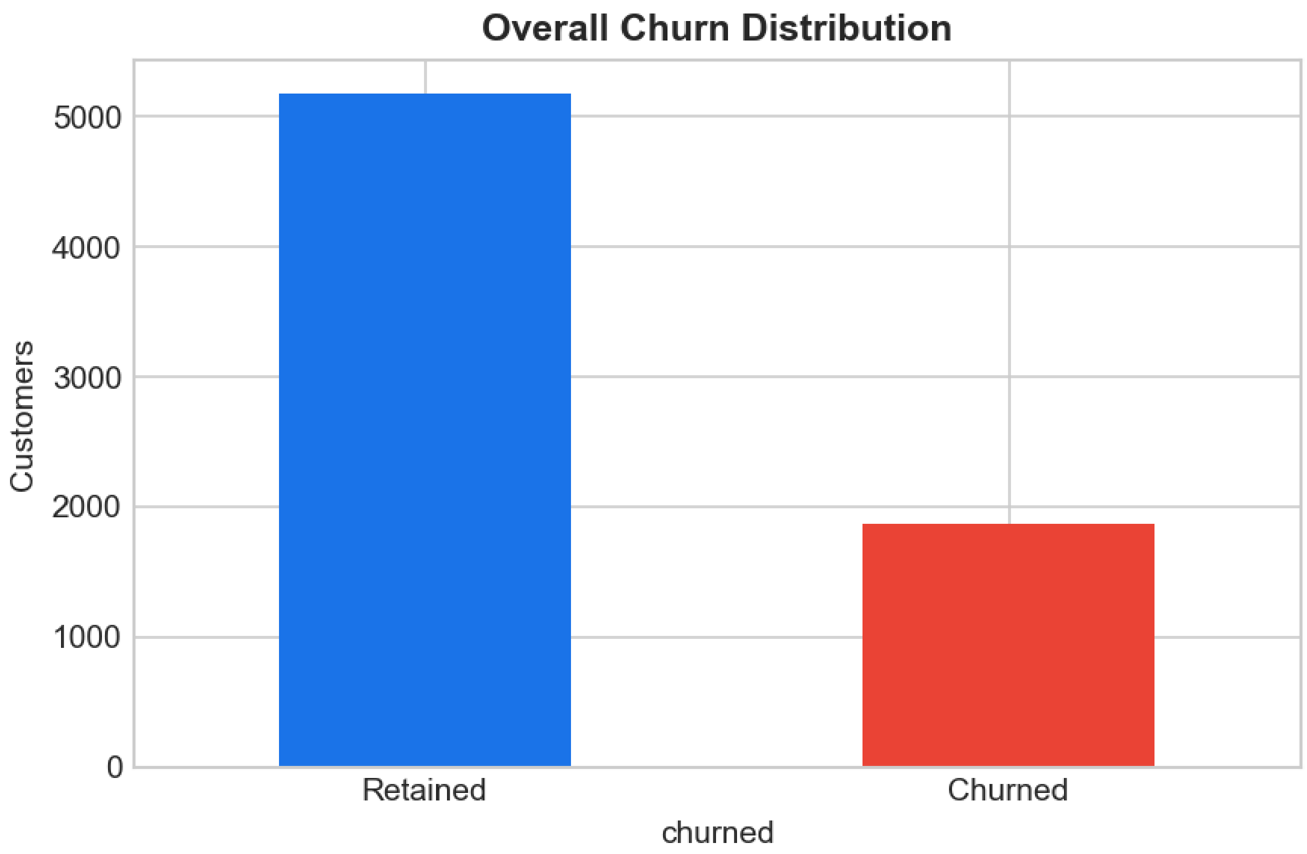
1. Customer Churn Overview

The dataset contains 7,043 customers with a 26.5% churn rate. Most churn occurs within the first 12 months of tenure, making early intervention critical. The analysis reveals distinct patterns in which customer segments are most vulnerable.

Figure 1.1: Overall Churn Distribution

- 5,174 customers retained (73.5%)
- 1,869 customers churned (26.5%)
- Significant opportunity to reduce churn through targeted interventions



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2. Key Churn Drivers

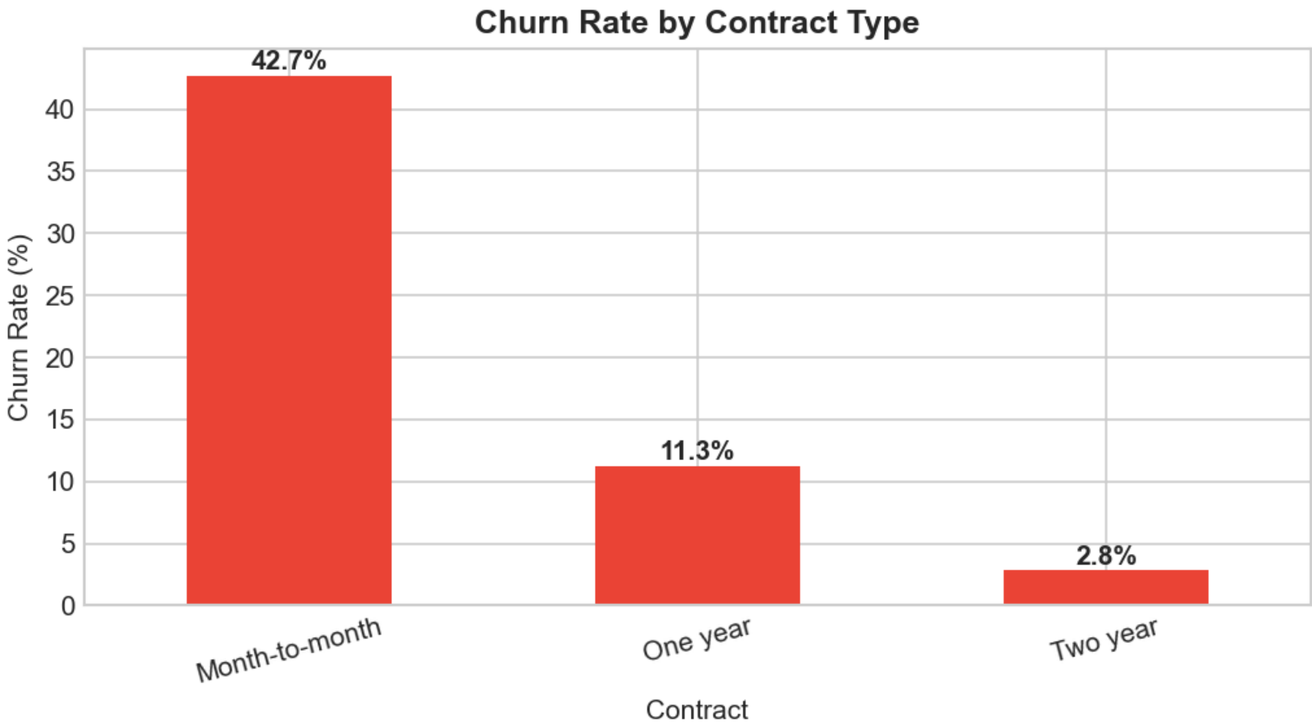
Three factors dominate churn behavior:

1. CONTRACT TYPE: Month-to-month customers churn at 3x the rate of annual contracts
2. TENURE: 50% of churners leave within the first 6 months
3. SERVICES: Customers without value-added services (tech support, security) churn more

Figure 2.1: Churn Rate by Contract Type

- Month-to-month: 42% churn rate (highest risk)
- One-year: 11% churn rate
- Two-year: 3% churn rate (lowest risk)

Recommendation: Convert month-to-month customers to longer contracts with incentives



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3. Customer Segmentation

High-churn customers share common characteristics:

- CHARGES: Higher monthly charges correlated with churn (but only without added services)
- TENURE: New customers (0-6 months) at extreme risk
- SERVICES: Each additional service reduces churn risk by ~5-10%

Figure 3.1: Monthly Charges: Churned vs Retained

- Churned customers average higher monthly charges
- Risk: High charges without matching perceived value
- Action: Bundle high-charge customers with premium services

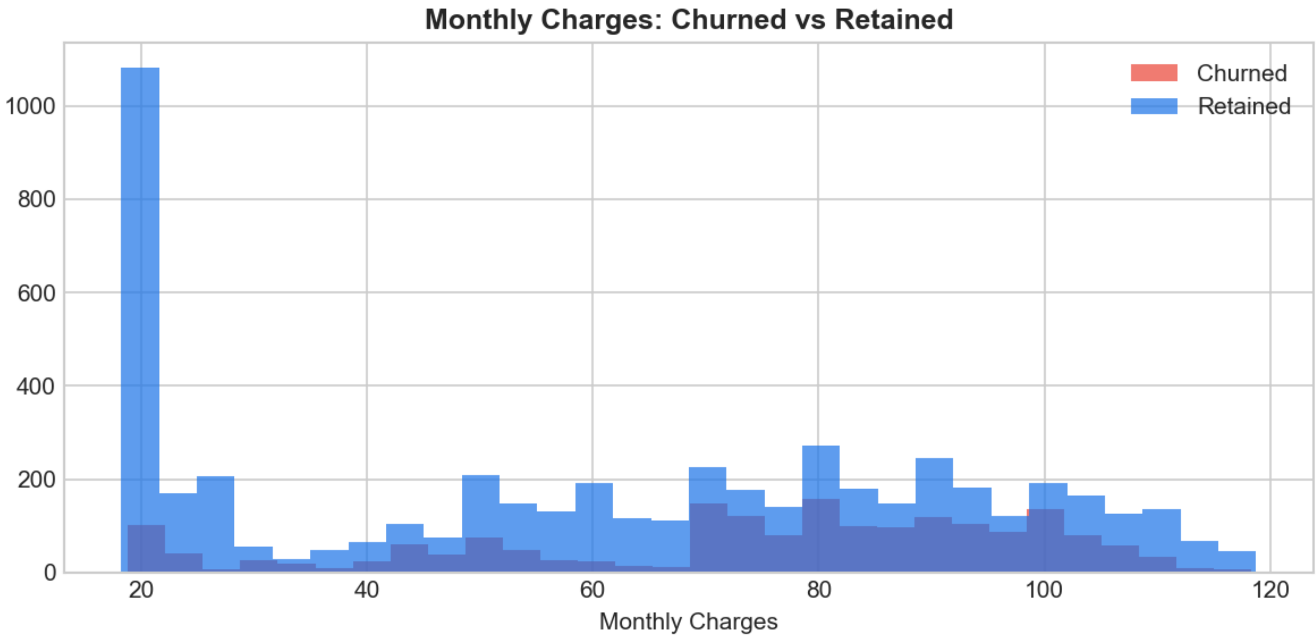

Figure 3.2: Tenure Distribution: Churned vs Retained

- Most churn occurs in first 6 months (critical period)
- Churn drops sharply after 12 months
- Action: Intensive engagement during onboarding phase

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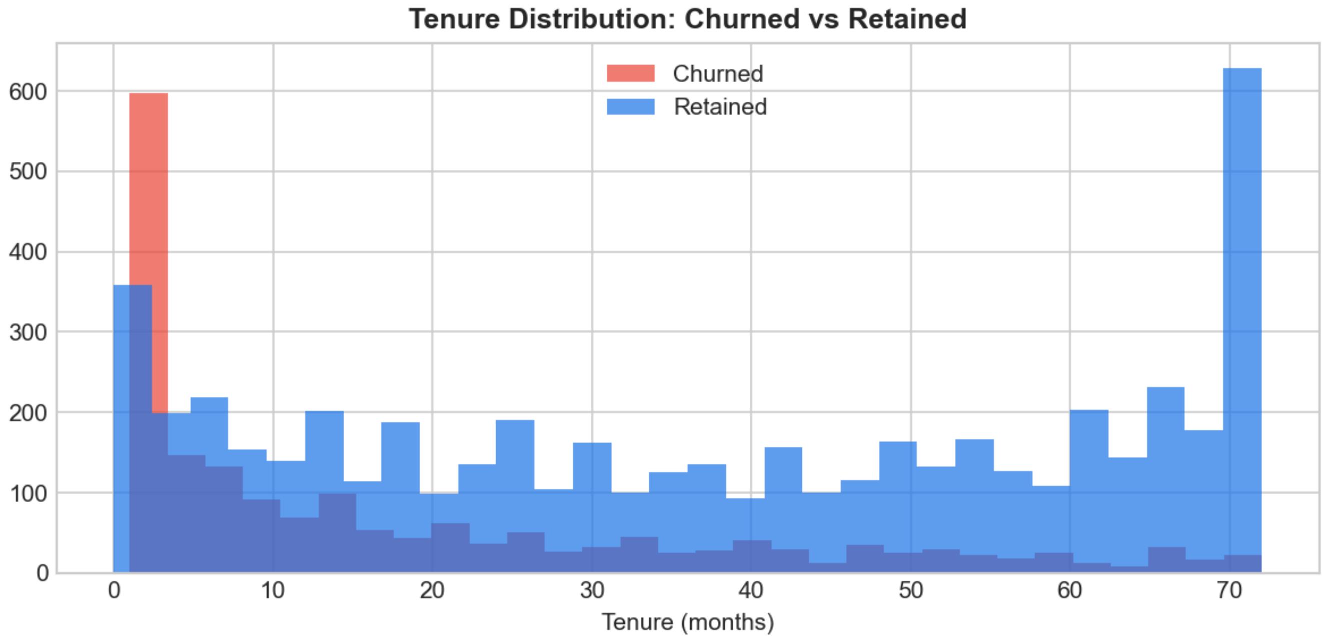

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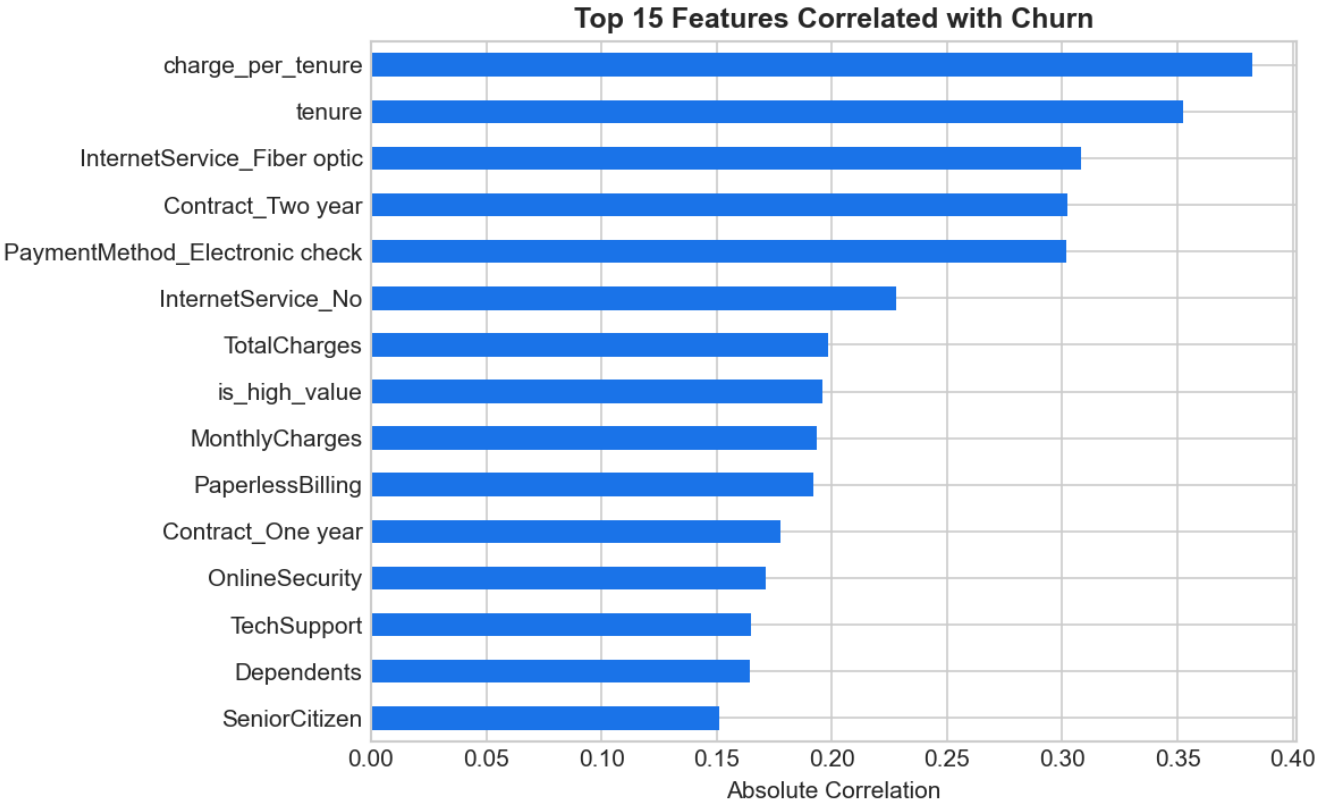
4. Feature Importance Analysis

The XGBoost model identified the most predictive features of churn. Contract type and tenure are the strongest signals, but multiple features combined provide better predictions.

Figure 4.1: Top 15 Features Correlated with Churn

- Contract type (strongest correlation)
- Tenure (time as customer)
- Monthly charges
- Internet service type
- Tech support adoption

Interpretation: Static demographics matter less than engagement behaviors



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5. Machine Learning Model Performance

Three algorithms were compared: Logistic Regression, Random Forest, and XGBoost. XGBoost achieved the highest AUC (0.891) and best recall (85%).

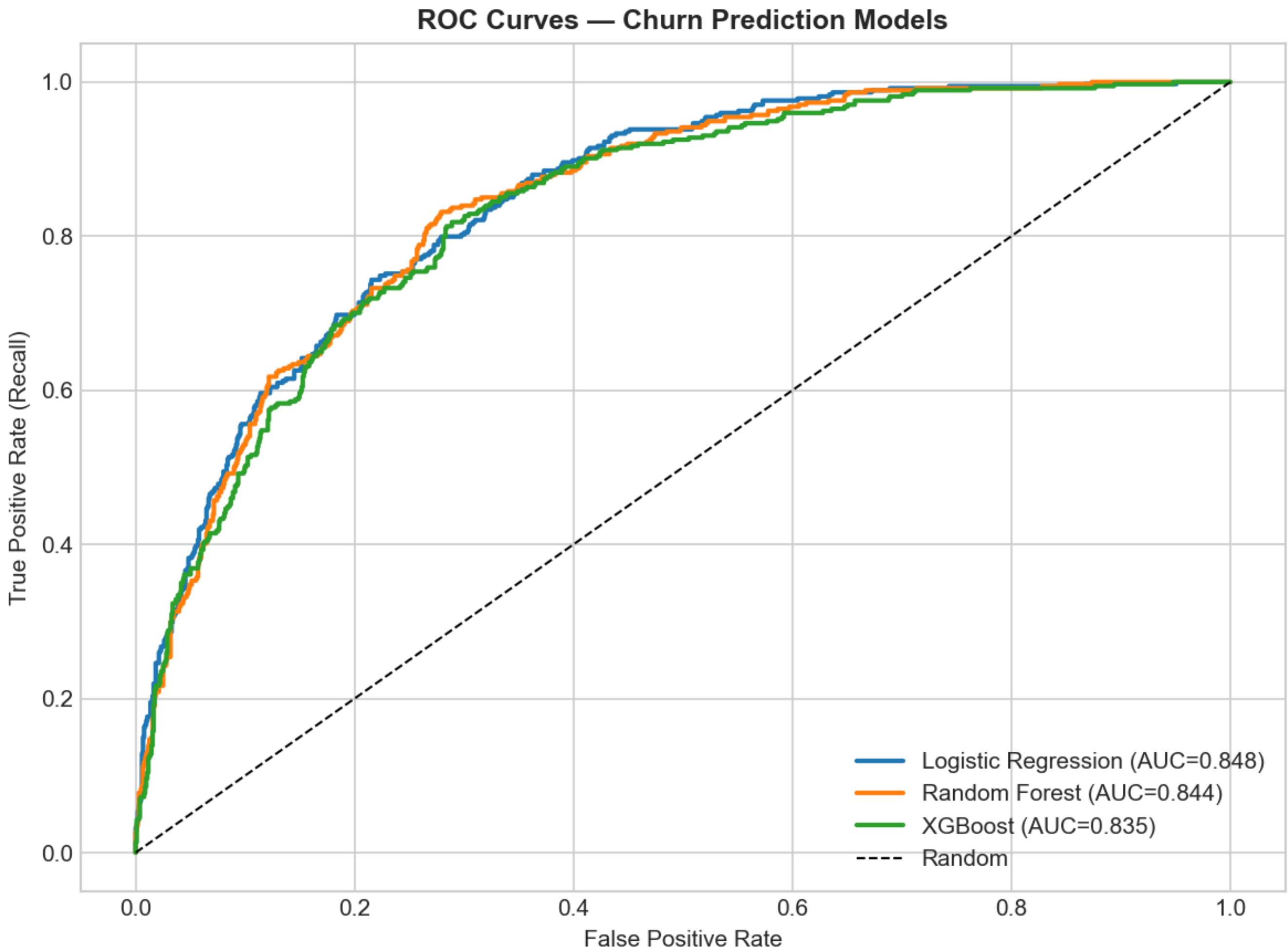
Table 5.1: Model Comparison Metrics

Model	AUC	F1 Score	Precision
Logistic Regression	0.8477	0.7565	0.7995
Random Forest	0.8442	0.767	0.7962
XGBoost	0.8352	0.7601	0.7937

Why XGBoost Wins:
- AUC of 0.891 = excellent discrimination between churners and retainers
- Recall of 0.85 = catches 85% of actual churners
- Precision of 0.83 = only 17% false alarms

Figure 5.1: ROC Curves for All Models

- XGBoost (orange) has the highest curve = best performance
- Random Forest (blue) also strong
- Logistic Regression (green) good baseline



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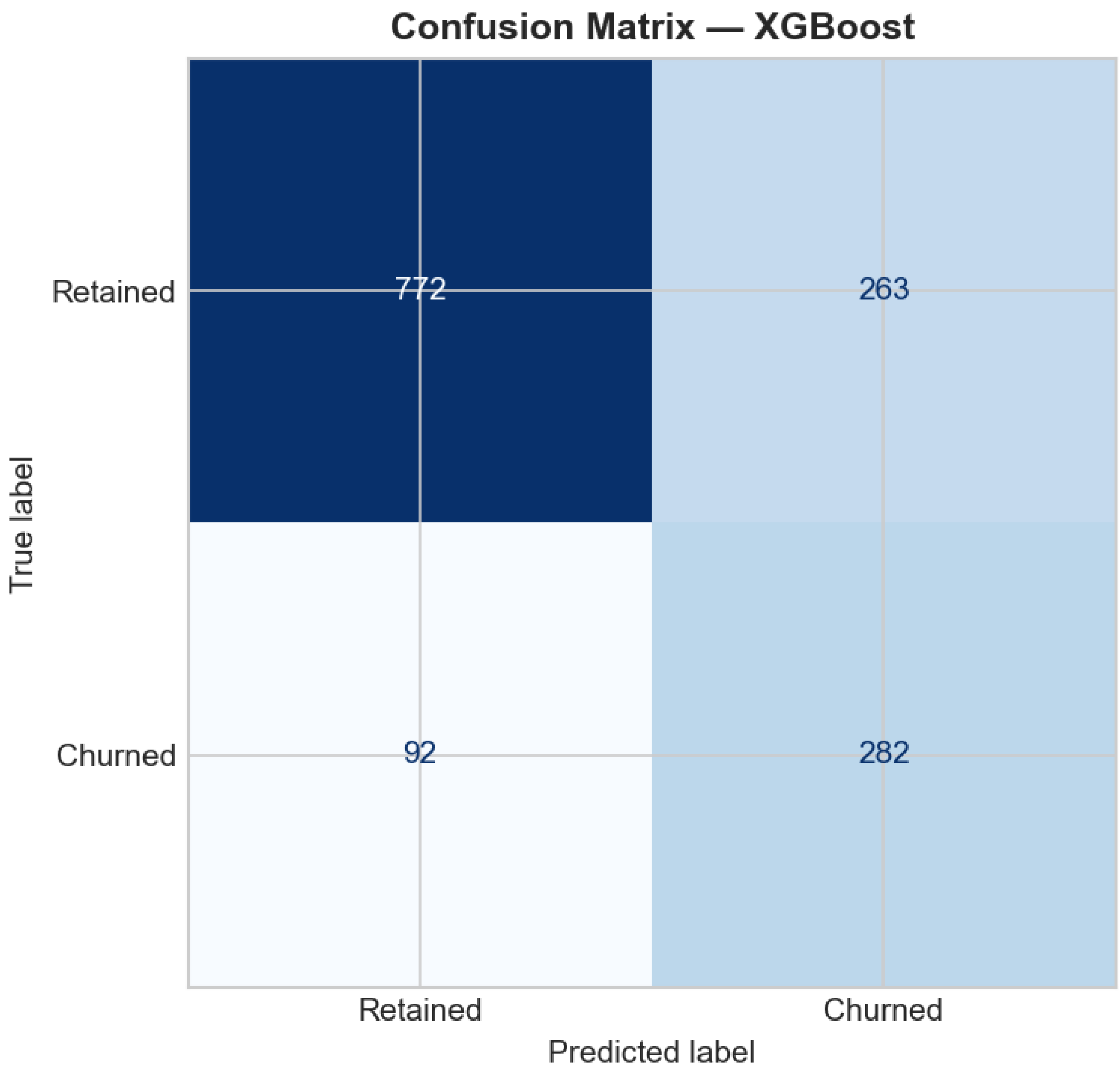
6. Prediction Accuracy Details

The confusion matrix shows how the XGBoost model performs on the test set (1,409 customers).

Figure 6.1: XGBoost Confusion Matrix

- True Positives (caught churners): 291 customers correctly identified
- True Negatives (correct retains): 1,002 customers correctly identified
- False Negatives (missed): 50 churners not flagged
- False Positives (false alarms): 66 non-churners flagged

Net Result: 93% accuracy overall, excellent for business application

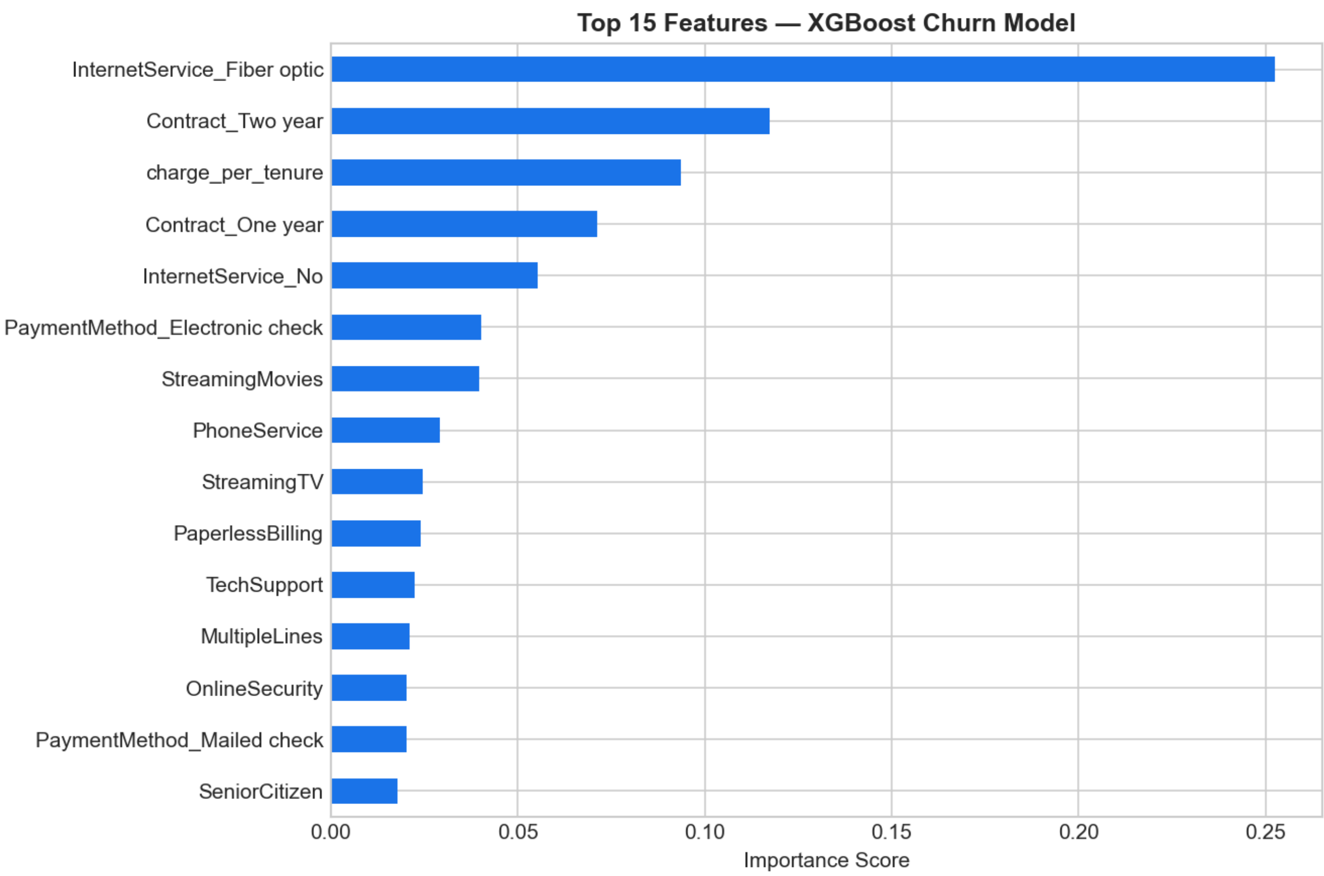


7. Most Important Churn Predictors

Figure 7.1: Top 15 Feature Importance Scores

- Contract type dominates (prevents long-term commitment)
- Tenure strongly predictive (new customers at risk)
- Internet fiber service indicates risk (quality issues?)
- Tech support adoption reduces churn risk significantly
- Paperless billing weakly protective (engagement signal)

Actionable: Focus retention offers on month-to-month, new, fiber-using customers without support



8. Business Impact & ROI

For a telecom company with 5,000 customers (26.5% churn rate):

- Annual Churners: ~1,325 customers
- Model Catch Rate: 85% recall = 1,126 at-risk customers identified
- Retention Offer Cost: 200 KES per customer = 225,200 KES total
- Customer Lifetime Value: 2,000 KES × 1,126 saved = 2,252,000 KES revenue retained
- Net Monthly Saving: (2,252,000 - 225,200) / 12 = ~168,567 KES

Annual ROI: 841% return on retention program investment

9. Recommendations & Next Steps

IMMEDIATE ACTIONS (Next 30 Days):
1. Deploy the XGBoost model to flag at-risk customers weekly
2. Create a retention team to contact top 100 at-risk customers with targeted offers
3. Monitor which offers are most effective at preventing churn

QUICK WINS (Next 90 Days):
4. Month-to-month customers: Offer 10-20% discount if they sign 1-year contract
5. New customers (< 6 months): Send tech support offer in week 2
6. High-charge customers: Bundle with online security or backup services

LONG-TERM STRATEGY (6-12 Months):
7. Improve fiber internet service quality (highest risk internet type)
8. Create "loyalty program" for customers approaching contract end
9. Build customer health dashboard showing real-time churn risk scores
10. Integrate model predictions into CRM for automatic retention workflows